Cut Set & Articulation Points

Dr. Sayantan Panja

1 Cut Set

Definition 1.1. *Let $G = (V, E)$ be a connected graph. A subset $F \subseteq E$ is called a **cut set** of G if the graph $G - F$ obtained by deleting all the edges in F is disconnected.*
i.e., in a connected graph G, the set of edges is said to be a cut set of G, if removal of the set makes the graph disconnected.

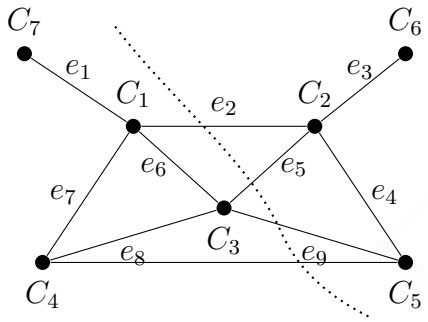


Fig. 1.1

In the above graph (Fig. 1.1) the set of edges $\{e_2, e_5.e_9, e_{10}\}$ is a cut set. Similarly, $\{e_2, e_5, e_4\}$ is also a cut set, $\{e_1\}$ is also a cut set.

Definition 1.2. *A cut set F is called a minimal cut set if no proper subset of F disconnects the graph. In other words, for every edge $e \in F$, the graph $G-(F\setminus\{e\})$ remains connected.*

We find that, $\{e_2, e_5.e_9, e_{10}\}$ is a cut set, but no proper subset of this set disconnectes the graph. Hence, this cut set is minimal cut set of the given graph(Fig. 1.1).

Note 1.3. *1. A cut set always cuts a graph into two.*

2. A cut set is a subgraph of G.

3. Every edge of a tree is a cut set of the tree.

Definition 1.4. *Let, T be a spanning tree of a connected graph G. A cut set S of G containing exactly one branch of T is called a fundamental cut set of G with respect of T.*

In the following graph (Fig. 1.2), T is a spanning tree of G shown by bold lines. The set of edges $\{BD, BC\}$ is a fundamental cut-set of G with respect to T, since it contains one branch BC of T.

In the same graph, the set $\{BD, CD, EF\}$ (shown by the dotted curve in the above figure) is a cut-set but not a fundamental cut-set with respect to T, because it contains two edges of the spanning tree T.

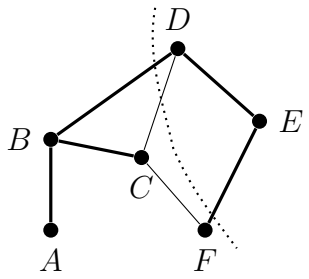


Fig. 1.2

2 Articulation Points or Cut Vertices

Definition 2.1. *A vertex v of a connected graph G is said to be an articulation point(or cut vertex) if its deletion from G (together with the edges incident to it) disconnects the graph.*

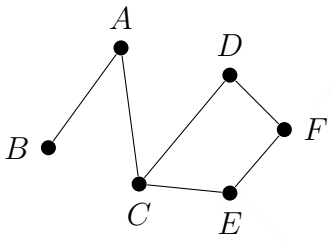


Fig. 2.1(a)

In the graph G shown in Fig.2.1(a), removal of the vertex A (together with the edges incident to it) leaves a disconnected graph shown in Fig. 2.1(b).

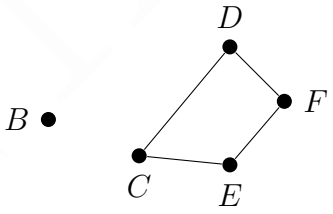


Fig. 2.1(b)

So, A is an articulation point. Similarly, C is also an articulation point, but F is not an articulation point of this graph.

Note 2.2. *1. A graph may have more than one articulation point.*

2. Every vertex (with degree greater than one) of a tree is a cut vertex.

3. A graph may have no cut vertex at all.

Example 2.3. *(i) Prove that a vertex of a tree is a cut vertex if and only if its degree is greater than* 1.

(ii) Prove that the number of cut vertices in a binary tree having n vertices is $\frac{n-1}{2}$.

Solution. **(i)** Let T be a tree and let v be a vertex of T such that $\deg(v) > 1$. Then v has at least two incident edges, say e_1 and e_2. Let the other ends of these edges be the vertices v_1 and v_2 respectively.

Then v_1 and v_2 are connected by the path $v_1 - v - v_2$ in T. By the property of a tree, this path is unique. So, if v (together with its incident edges) is removed from T, the vertices v_1 and v_2 become disconnected. Therefore v is a cut vertex of T.

Conversely, let v is a cut vertex of a graph G. Then removing v produces more than one component. Let G_1 and G_2 be two such components containing vertices v_1 and v_2 respectively.

In the original graph G, the edges vv_1 and vv_2 must exist. These are two distinct edges incident to v. Hence $\deg(v) > 1$.

Therefore, a vertex of a tree is a cut vertex if and only if its degree is greater than 1.

(ii) Let T be a binary tree having n vertices. Then T has $\frac{n+1}{2}$ pendant vertices. That is, T has $\dfrac{n+1}{2}$ vertices of degree 1.

Hence the number of vertices whose degree is greater than 1 is $n - \frac{n+1}{2} = \frac{n-1}{2}$.

From part (i), every vertex whose degree is greater than 1 is a cut vertex. Therefore, the number of cut vertices in T is $\frac{n-1}{2}$.

Example 2.4. *For the following graph, indicate the cut vertices and the vertices which are not cut vertices. Display the graph obtained after removal of the cut vertex.*

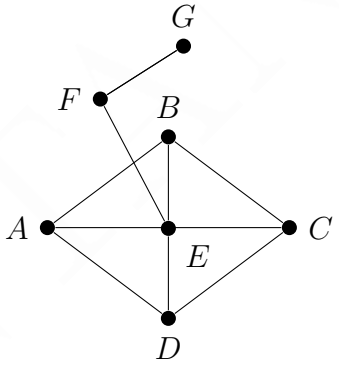


Fig. 2.2

Example 2.5. *For the following graph, find a cut set with respect to the vertices C and D.*

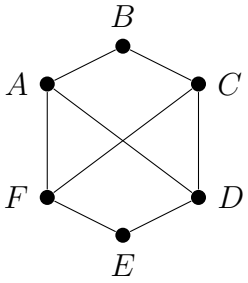


Fig. 2.3